A machine-checked formalization of the Fuxi hexagram system: group structure, divination-induced Markov dynamics, and corrections to the received analysis

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Abstract

The Fuxi Earlier Heaven arrangement of the sixty-four hexagrams has been read as a binary system since Leibniz, and a series of algebraic and coding-theoretic studies have since described parts of its structure. Those descriptions are stated informally, and their quantitative claims have not been checked against a running model. We give a complete computational formalization of the system and an open-source suite that machine-checks every quantitative claim it makes. We first fix a bottom-up bit encoding and show by exhaustion that the recursive doubling generation produces the full set of six-bit strings under either of the two line-placement conventions in circulation. We then prove that the changing-line mechanism is bitwise exclusive-or with a mutation mask, and identify the resulting transition structure as the elementary abelian group $\mathbb{Z}_2^6$, whose Cayley graph is the six-dimensional hypercube. Extending the model to a Markov chain exposes a discrepancy that informal treatment had hidden. Under the yarrow-stalk procedure a yang line changes with probability 3/8 and a yin line with probability 1/8, so the mutation mask is not independent of the current state even though its marginal rate is 1/4. The kernel the procedure actually induces is therefore asymmetric, and its stationary distribution is not uniform but the product form $\pi(h) \propto (3/4)^{\text{yin}}(1/4)^{\text{yang}}$, which places 729 times more mass on the all-yin hexagram than on the all-yang hexagram. Of forty checked claims, twenty-four are confirmed, five are refined, four are new, and seven are corrected. The corrections include a clustering coefficient reported as 5/12 that is exactly zero because the state graph is bipartite, a mutual information of 1.132 bits that is 1.233 bits once the state dependence is carried through, and a genetic-code dimension count that overstates the independent binary degrees of freedom by a factor of two per nucleotide. The results are reproducible from a single command, and we state explicitly which conclusions are formal theorems, which are exhaustive machine checks, and which are estimates under a stated model of how the divining pile is divided.

Keywords: Fuxi hexagram arrangement; binary encoding; elementary abelian group; hypercube; Markov chain; yarrow-stalk divination; reproducible verification

1 Introduction

The sixty-four hexagrams of the *Yijing* are six-line figures built from two kinds of line. In 1703 Leibniz published an account of binary arithmetic that explicitly connected his notation to hexagram diagrams sent from China by the Jesuit Joachim Bouvet [1]. The correspondence between the two line types and the digits 0 and 1 has been revisited many times since, and the Fuxi Earlier Heaven arrangement in particular has been analysed as an algebraic object [3, 5, 6] and as a code [4].

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These treatments share a methodological gap. They are stated as prose and hand-derived arithmetic, and their numerical claims have not been executed. That is a problem for two reasons. First, several of the quantities involved are easy to state and hard to compute by inspection, including graph invariants of a 64-vertex graph and entropies of a joint distribution over four line states. Second, the system carries a long interpretive tradition, which makes it unusually easy for a claim to be repeated because it sounds structurally plausible rather than because it has been checked.

The practical consequence is that the received description of the system contains errors that survive precisely because nobody has run it. A clustering coefficient is reported for the state-transition graph, and the graph in question is bipartite and therefore has none. A Markov chain is derived from the yarrow-stalk divination procedure under an independence assumption that the procedure does not satisfy. A count of binary degrees of freedom in the genetic code is used to reach the number 64 by a route that would in fact give 512.

This paper closes that gap. We give a formalization that is complete enough to execute, and an accompanying suite that checks every quantitative claim the formalization makes. Our contributions are the following.

1. We fix a bottom-up bit encoding and show by exhaustive construction that the recursive doubling generation attributed to Shao Yong yields exactly the set of six-bit strings, under either of the two conventions for where a new line is placed. This makes precise the sense in which the binary reading does not depend on reading direction (Section 3.1).

2. We prove that the changing-line mechanism is bitwise exclusive-or with a mutation mask, and verify the proof on all 4^6 assignments of the four traditional line states. We then identify the transition structure as the elementary abelian group $\mathbb{Z}_2^6$. The three algebraic properties that earlier treatments list separately are the group axioms, and the state graph is the Cayley graph of that group (Section 3.2).

3. We model the yarrow-stalk procedure exactly and show that it induces a transition kernel different from the one usually assumed. A yang line changes three times as often as a yin line, so the mutation mask is state dependent. The induced kernel is asymmetric and its stationary distribution is not uniform (Sections 3.3 and 3.4).

4. We compute the topological and information-theoretic quantities directly from the model rather than from quoted formulas, and report where the computed values differ from the received ones (Section 4).

5. We release the verification suite. Every number in this paper is produced by a single command, and each check records whether it confirms, refines, corrects, or adds to the prior account (Section 3.6).

We are deliberately narrow about what these results mean. The system is a six-bit finite state machine. Nothing here bears on the interpretive or divinatory tradition, and nothing here supports the claim that the arrangement anticipated any later scientific development. Section 5.3 states the boundary explicitly, because the surrounding literature does not always do so.

2 Related work

Historical and philological readings. The transmission of the diagrams to Leibniz through Bouvet, and the subsequent reconstruction of that exchange, are documented by Aiton and Shimao [1]. A recurring question is whether the binary reading is intrinsic to the arrangement or projected onto it, since Leibniz assigned the highest bit weight to the bottom line and later authors have not. Maitre argues that the binary structure follows from the generation method itself and is therefore not an artifact of reading direction [2]. That argument is about the diagram; it does not supply an executable model, and it leaves the enumeration order unaddressed. We verify the set-level claim and extend it to the ordering in Section 3.1.

Algebraic treatments. Goldenberg analysed hexagram transformations group-theoretically and reported a non-abelian structure [3]. Schöter developed a Boolean-algebra framework covering correspondence and bipolar change [5, 6]. These works operate on the same underlying set as we do, and the difference is which group action is in view. Two distinct groups act on the sixty-four hexagrams. Permuting the six line positions gives S_6, which is non-abelian. Flipping line values gives the elementary abelian group $\mathbb{Z}_2^6$. The changing-line mechanism lives entirely in the second, which is why the structure we describe in Section 3.2 is abelian and is not in conflict with a non-abelian result about a different action. We make the separation explicit and compute the order of the group the two generate together.

Coding-theoretic treatments. Hayata computed Hamming distances between consecutive hexagrams under several orderings and mapped the hexagrams onto a six-dimensional hypercube [4]. That work establishes the hypercube embedding, which we take as given, and it reports distance statistics rather than graph invariants. The clustering coefficient we correct in Section 4.5 is therefore not a claim of that paper. It is a value carried in the informal restatement of the system that this work formalizes, and we have not traced it to a primary source.

Probabilistic models of the divination procedure. The line-state probabilities of the yarrow-stalk method have been derived combinatorially and checked by simulation [10, 11]. Those studies compute the marginal distribution over the four line states, which is what the procedure directly produces. They do not ask what transition kernel the procedure induces on the space of hexagrams. That question is where the independence assumption enters, and it is where we find the discrepancy reported in Section 3.4.

Structural comparisons with the genetic code. Both the hexagram set and the codon set have 64 elements, and this has motivated a body of comparative work [7, 8, 9]. The comparisons rest on a count of binary dimensions per nucleotide. We show in Section 4.7 that the three dichotomies usually cited are not independent, which changes the count although it does not change the conclusion that both sets have 64 elements.

3 Formal model

3.1 Encoding and the doubling method

A hexagram is six stacked lines, each either yang or yin. Traditional line names run from the bottom upward. We write a hexagram as a tuple $(b_1, \ldots, b_6) \in \{0,1\}^6$ with b_1 the bottom line, and set $b_i = 1$ for yang and $b_i = 0$ for yin.

We adopt the bottom-up bit-weight convention, in which the bottom line is the least significant bit:

$$V(h) \;=\; \sum_{i=1}^{6} b_i \, 2^{\,i-1}. \tag{1}$$

Under Equation (1) the all-yin hexagram Kun is 0 and the all-yang hexagram Qian is 63, and V is a bijection onto $\{0, \ldots, 63\}$.

The recursive doubling generation attributed to Shao Yong starts from a single undifferentiated symbol and splits every symbol in two at each step. Writing $G(n)$ for the set of n-line symbols, $G(0)$ is a singleton and each step replaces every symbol by two symbols differing in one new line.

Theorem 1. $|G(n)| = 2^n$, *and* $G(6)$ *is the full set of six-bit strings.*

Proof. By induction. $|G(0)| = 1 = 2^0$. Each element of $G(k)$ produces exactly two elements of $G(k+1)$, and the two are distinguished by the value of the new line, so the branches are disjoint and $|G(k+1)| = 2|G(k)|$. Since the construction never identifies two distinct symbols, $G(n)$ injects into $\{0,1\}^n$, and equality of cardinalities gives surjectivity. □

Theorem 1 is convention-independent in a sense worth making precise, because the placement of the new line is exactly the point on which readings differ. The new line may be added above the existing ones, becoming the new most significant bit, or below them, shifting the existing lines up. We verify in Section 4 that both placements generate the same set and, with the natural enumeration at each level, the same order.

The set carries a Boolean lattice structure under bitwise implication, with meet given by bitwise AND, join by bitwise OR, bottom element Kun, and top element Qian. The rank function is the number of yang lines, so the rank distribution is $\binom{6}{k}$ for $k = 0, \dots, 6$.

3.2 Changing lines as exclusive-or

The divination assigns each line one of four states: old yin (6), young yang (7), young yin (8), and old yang (9). The two old states change into their opposites and the two young states are stable. A cast therefore determines two hexagrams, which we call the cast hexagram and the transformed hexagram.

Let $M = (m_1, \dots, m_6)$ be the mutation mask, with $m_i = 1$ exactly when line i is old.

Theorem 2. *The transformed hexagram is* $B' = B \oplus M$, *where* $\oplus$ *is bitwise exclusive-or.*

Proof. Fix a line i. If $m_i = 0$ the line is young and $b'_i = b_i = b_i \oplus 0$. If $m_i = 1$ the line is old and changes, so $b'_i = 1 - b_i = b_i \oplus 1$. In both cases $b'_i = b_i \oplus m_i$, and the claim follows coordinatewise. □

Theorem 2 lets the system be presented as a deterministic finite automaton with state set $Q = \{0, \dots, 63\}$, input alphabet $\Sigma = \{0, \dots, 63\}$, and transition function $\delta(q, m) = q \oplus m$. Earlier accounts then list three properties of δ: transitions commute, each transition is its own inverse, and any state reaches any other. We verify all three, and note that they are not three facts.

Proposition 3. $(Q, \oplus)$ *is the elementary abelian group* $\mathbb{Z}_2^6$. *The three listed properties are, respectively, commutativity, the involution property of an exponent-two group, and the transitivity of the regular action. The state-transition graph under single-line changes is the Cayley graph of* $\mathbb{Z}_2^6$ *with respect to the six standard generators, which is the six-dimensional hypercube* Q_6.

Presenting the structure as a group has two consequences we use later. It predicts the graph invariants of Section 4.5 without further computation, and it separates cleanly from the other natural group action. Flipping line values gives $\mathbb{Z}_2^6$, while permuting the six line positions gives S_6. The two combine as a semidirect product, the hyperoctahedral group of order $2^6 \cdot 6! = 46{,}080$. The changing-line mechanism lives entirely in the normal subgroup $\mathbb{Z}_2^6$.

3.3 The yarrow-stalk procedure as a probability model

The classical procedure produces one line through three operations on a pile of 49 stalks. At each operation the pile is divided into two heaps, one stalk is taken from the right heap, each heap is counted off in fours, and the remainders together with the single stalk are set aside. After three operations the remaining pile divided by four is 6, 7, 8, or 9.

The textbook analysis assumes the left-heap size is uniform modulo four. Under that assumption the first operation sets aside 5 with probability 3/4 and 9 with probability 1/4, and the second and third set aside 4 or 8 with probability 1/2 each. Propagating these gives

$$P(6) = \tfrac{1}{16}, \quad P(7) = \tfrac{5}{16}, \quad P(8) = \tfrac{7}{16}, \quad P(9) = \tfrac{3}{16}. \tag{2}$$

We implement the procedure exactly and evaluate Equation (2) in rational arithmetic, and we also evaluate a model in which the split point rather than its residue class is uniform. The uniformity assumption holds exactly only when the number of admissible split points is divisible by four, which is true at the first operation and false at the second and third. Section 4 reports how far the resulting probabilities move.

The step that matters is the following reading of Equation (2). The four line states are not merely a distribution over outcomes. They are the joint distribution of the pair (cast bit, transformed bit):

$$\begin{aligned} P(\text{yin}, \text{yin}) &= P(8) = \tfrac{7}{16}, & P(\text{yin}, \text{yang}) &= P(6) = \tfrac{1}{16}, \\ P(\text{yang}, \text{yin}) &= P(9) = \tfrac{3}{16}, & P(\text{yang}, \text{yang}) &= P(7) = \tfrac{5}{16}. \end{aligned} \tag{3}$$

Reading the states as a joint law makes the conditional flip probabilities visible:

$$P(\text{change} \mid \text{yang}) = \frac{P(9)}{P(7) + P(9)} = \frac{3}{8}, \qquad P(\text{change} \mid \text{yin}) = \frac{P(6)}{P(6) + P(8)} = \frac{1}{8}. \tag{4}$$

The marginal change probability is $P(6) + P(9) = 1/4$, and the expected number of changing lines is 3/2. Both match the received values. The two conditionals in Equation (4) differ by a factor of three, and it is this that the received analysis does not carry through.

3.4 Two transition kernels

We compare two kernels on the 64 states.

The *independent-mask kernel* P^A is the one assumed in the received analysis. Each line flips independently with probability $p = 1/4$ regardless of its current value, so

$$P^A_{ij} = p^{\,w(i \oplus j)} (1 - p)^{\,6 - w(i \oplus j)}, \tag{5}$$

where w is the Hamming weight. This kernel depends on i and j only through $i \oplus j$. It is therefore symmetric, and a symmetric stochastic matrix has the uniform distribution as a stationary distribution.

The *yarrow kernel* P^B is the one the procedure induces. Using Equation (4), line k flips with probability $q(b_k)$ where $q(1) = 3/8$ and $q(0) = 1/8$, so

$$P^B_{ij} = \prod_{k=1}^{6} \begin{cases} q(b_k) & \text{if line } k \text{ differs between } i \text{ and } j, \\ 1 - q(b_k) & \text{otherwise,} \end{cases} \tag{6}$$

with b_k the kth line of i. Because $q(1) \neq q(0)$, this is not a function of $i \oplus j$, and P^B is not symmetric.

Proposition 4. *The stationary distribution of P^B is the product form*

$$\pi(h) \;=\; \left(\tfrac{1}{4}\right)^{y(h)} \left(\tfrac{3}{4}\right)^{6 - y(h)}, \tag{7}$$

where $y(h)$ is the number of yang lines of h.

Proof. The six lines are cast independently, so P^B is a sixfold tensor power of the two-state kernel $\left(\begin{smallmatrix} 7/8 & 1/8 \\ 3/8 & 5/8 \end{smallmatrix}\right)$ written in the order (yin, yang). Detailed balance for a two-state chain gives $\pi(\text{yin})/\pi(\text{yang}) = q(1)/q(0) = 3$, so the per-line stationary distribution is $(3/4, 1/4)$. The stationary distribution of a tensor power is the tensor power of the stationary distribution, which is Equation (7). □

Equation (7) is far from uniform. It assigns $\pi(\text{Kun}) = (3/4)^6 = 729/4096$ and $\pi(\text{Qian}) = (1/4)^6 = 1/4096$, a ratio of 729 to 1. The chain drifts toward yin because yang lines are the ones that change more readily.

Both kernels are sixfold tensor powers of a two-state kernel with second eigenvalue 1/2, so both have eigenvalues 2^{-k} with multiplicity $\binom{6}{k}$ and spectral gap 1/2. The correction to the stationary distribution therefore leaves the mixing rate intact.

3.5 Information content

Under a uniform prior each hexagram carries $\log_2 64 = 6$ bits. For the independent-mask model with B uniform and M independent of B, the transformed hexagram $B' = B \oplus M$ is again uniform and

$$I^A(B; B') = H(B') - H(B' \mid B) = 6 - 6\, H_b(1/4), \tag{8}$$

where H_b is the binary entropy function.

For the yarrow model the mutual information must be computed from the joint law in Equation (3) rather than from an assumed mask. Since the six lines are independent,

$$I^B(B; B') = 6 \cdot I(b; b'), \tag{9}$$

with $I(b; b')$ the mutual information of the 2×2 table in Equation (3). The transformed hexagram is yang on any given line with probability $P(6) + P(7) = 3/8$, so it is not uniform and carries less than six bits.

3.6 The verification suite

The formalization is implemented as a seven-module Python package with no required dependencies beyond the standard library. Rational quantities are computed with exact arithmetic, so probabilities, stationary distributions, and kernel entries are compared for exact equality rather than within a tolerance. NumPy is used only where a numerical eigendecomposition provides an independent cross-check of an analytically predicted spectrum.

The driver evaluates forty claims and assigns each one of four verdicts. Confirmed means the computed value matches the value stated in the received account. Refined means the stated value holds under an idealization, which the check then makes explicit and quantifies. Corrected means the computed value contradicts the stated value. New means the quantity was not previously reported. Wherever a property is finite enough to enumerate, the check enumerates it rather than sampling. The exclusive-or equivalence of Theorem 2 is checked on all $4^6 = 4096$ line-state assignments, commutativity on all $64^3 = 262{,}144$ triples, and the graph invariants on the whole graph.

4 Verification results

Table 1 summarizes the outcome. Of forty checks, twenty-four are confirmed, five refined, seven corrected, and four new. We report the corrections in full, since they are the results that change the received account.

Table 1: Outcome of the forty checks, by area. The full report with per-check evidence is generated by the suite.

Area	Confirmed	Refined	Corrected	New	Total
Encoding and generation	5	1	0	0	6
Automaton and group	4	1	0	1	6
Yarrow-stalk procedure	5	1	2	1	9
Markov kernels	2	1	1	1	5
Topology	4	1	1	1	7
Information	3	0	2	0	5
Genetic code	1	0	1	0	2
Total	24	5	7	4	40

4.1 Encoding and generation

The doubling method returns 2^n symbols for every n from 0 to 6, and the set it returns at $n = 6$ equals the set of all six-bit strings computed independently. Both line-placement conventions pass. Adding the new line above the existing ones, with the yin branch of each level emitted first, gives the sorted sequence $0, 1, \ldots, 63$. Adding it below, with each parent emitting its yin child then its yang child, gives the same sorted sequence. The binary structure and the enumeration order therefore both survive the change of convention, which strengthens the set-level claim of [2] (Figure 1).

The Boolean-lattice rank distribution is $1, 6, 15, 20, 15, 6, 1$ as expected.

4.2 Algebraic structure

Theorem 2 holds on all 4096 assignments of the four line states, with zero mismatches against a line-by-line simulation of the traditional semantics. Commutativity holds on all 262,144 triples and the involution property on all 4096 pairs. For every ordered pair of states there is exactly one input symbol carrying the first to the second, so the automaton reaches any state from any other in one input step.

The group identification of Proposition 3 is confirmed: the structure is abelian of order 64 and exponent 2. Extending to permutations of line positions, the action of the generated group is faithful and has order $46{,}080 = 2^6 \cdot 6!$, matching the hyperoctahedral group. This quantity is not in the received account.

4.3 Divination probabilities and the failure of independence

Exact rational evaluation of the three-operation procedure reproduces Equation (2) precisely, as rationals rather than as decimal approximations. A Monte-Carlo simulation of the physical procedure over 10^6 casts agrees with the exact values to within 2.7×10^{-4} (Figure 2). The marginal change probability is exactly $1/4$ and the expected number of changing lines is exactly $3/2$, both matching the received values. The cast hexagram is uniform over the 64 states, since each line is yang with probability exactly $1/2$, which justifies the uniform prior used in the entropy calculation.

The correction is in the conditionals. Equation (4) gives $P(\text{change} \mid \text{yang}) = 3/8$ and $P(\text{change} \mid \text{yin}) = 1/8$. The mutation mask is therefore not independent of the state. The marginal rate of $1/4$ is a mixture of these two conditionals weighted by the equal marginals of yang and yin, which is why the dependence is invisible at the level of the marginal and why the received analysis could carry an independence assumption without an obvious inconsistency.

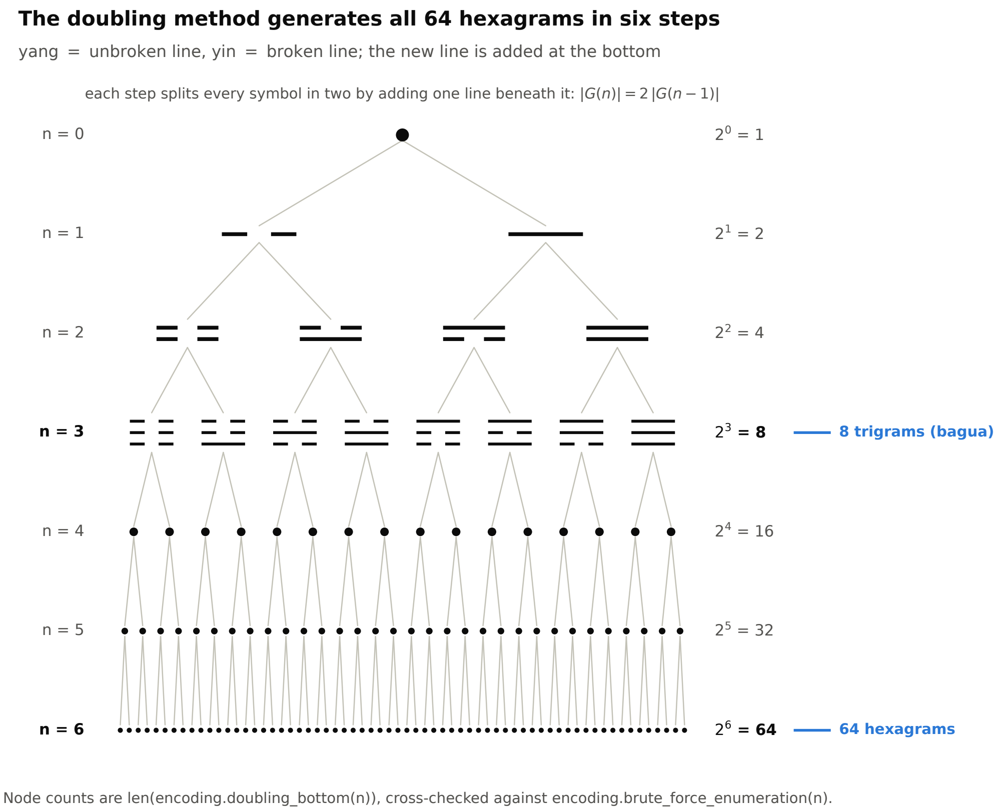

Figure 1: The recursive doubling generation. Each level doubles the number of symbols, giving the eight trigrams at level three and the sixty-four hexagrams at level six. The generated set is the same under either convention for placing the new line.

The transformed hexagram is yang on any given line with probability 3/8, not 1/2, so it is not uniform.

Under the uniform-split model the four line-state probabilities move by at most 1.3×10^{-2}, and the conditionals become 0.400 and 0.112. The idealized values are therefore a good approximation, and the state dependence is robust to the modelling choice rather than an artifact of it.

4.4 Stationary behaviour of the two kernels

The independent-mask kernel of Equation (5) is stochastic, symmetric, irreducible, and aperiodic, and the uniform distribution is exactly stationary for it. The received Theorem on this point is correct for the kernel it defines.

The yarrow kernel of Equation (6) is stochastic, irreducible, and aperiodic, but it is not symmetric, and the uniform distribution is not stationary for it. The product form of Equation (7) is exactly stationary, verified in rational arithmetic against the full 64×64 kernel. It places 729/4096 of the mass on Kun and 1/4096 on Qian (Figure 3).

Both kernels have spectral gap 1/2, agreeing with the analytically predicted eigenvalues 2^{-k} with multiplicity $\binom{6}{k}$. Measured worst-case mixing to total-variation distance 0.01 takes seven steps for the independent-mask kernel and eight for the yarrow kernel (Figure 4). The received account gives an order-of-magnitude bound of six steps, which the measurement refines.

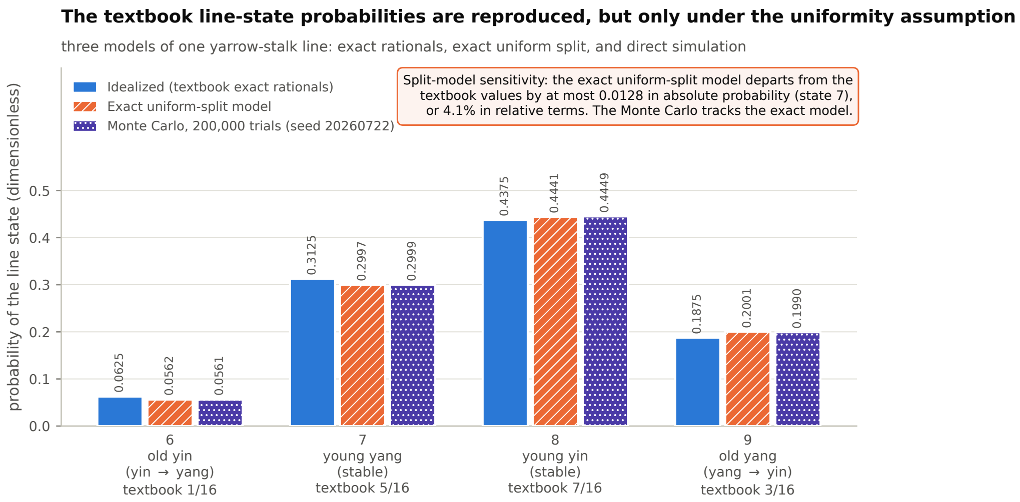

Figure 2: Line-state probabilities under three models. The idealized model reproduces the received values exactly. The uniform-split model quantifies the sensitivity of those values to the assumption that the left-heap size is uniform modulo four. The Monte-Carlo estimate confirms the idealized values.

4.5 Topology

The state-transition graph has 64 vertices, 192 edges, and is 6-regular. Its diameter is 6, measured by breadth-first search from all sources, and the graph distance equals the Hamming distance for every pair. The adjacency spectrum computed numerically agrees with the analytic prediction of $6 - 2k$ with multiplicity $\binom{6}{k}$.

Two quantities require amendment.

The clustering coefficient is exactly zero, not 5/12. The graph contains no triangles at all: enumeration over 960 connected triples returns zero closed ones. The reason is structural. The graph is bipartite, with the two parts given by the parity of the number of yang lines and each of size 32. A bipartite graph has no odd cycles and therefore no triangles. Equivalently, two neighbours of a hexagram each differ from it in one line, so they differ from each other in two lines and are never adjacent. Every clustering coefficient of Q_6 is zero, and no value in $(0, 1]$ can be correct (Figure 5).

The average shortest-path length is $192/63 \approx 3.048$, not 3. The value 3 is the mean Hamming distance between two independently uniform hexagrams, which counts the 64 zero-distance self-pairs. The standard definition of average shortest-path length excludes them. Both quantities are meaningful and they are not the same, so we report both.

The Hamming distance distribution between two independent uniform hexagrams has mean exactly 3 and variance exactly 3/2, matching the received values.

4.6 Information content

The entropy of a uniformly distributed hexagram is 6 bits. The entropy of the independent mutation mask at $p = 1/4$ is $6H_b(1/4) = 4.8677$ bits, and Equation (8) gives $I^A = 1.1323$ bits. These match the received values, and the arithmetic of the received calculation is correct for the model it assumes.

Carrying the state dependence through changes the answer. Equation (9) evaluated on the joint law of Equation (3) gives $I^B = 1.2326$ bits. The transformation retains 20.5 percent of the six bits rather than 18.9 percent. Relatedly, the transformed hexagram carries 5.7266 bits, not 6, because it is yin-biased (Figure 6).

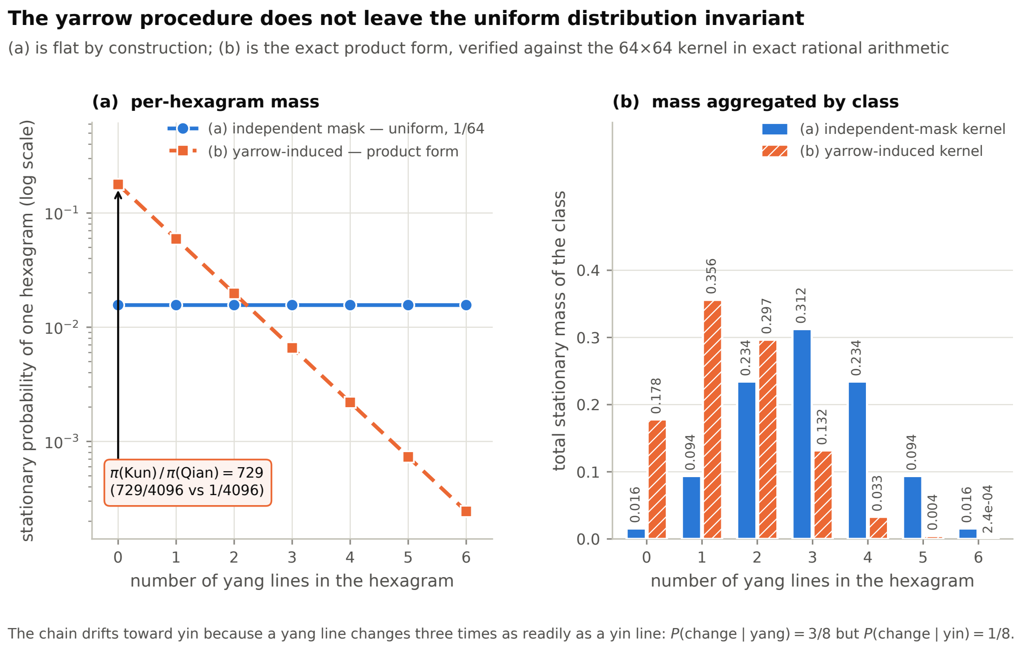

Figure 3: Stationary mass by number of yang lines. The independent-mask kernel is uniform over the sixty-four states. The kernel induced by the yarrow-stalk procedure is not, because yang lines change more readily than yin lines. The ratio between the all-yin and all-yang states is 729 to 1.

4.7 The genetic-code dimension count

Each nucleotide is commonly described by three binary dichotomies: purine against pyrimidine, amino against keto, and strong against weak. Applying three dichotomies at three codon positions suggests nine binary degrees of freedom, which would give $2^9 = 512$ codons rather than 64.

Gaussian elimination over GF(2) on the 4×3 dichotomy matrix returns rank 2, not 3. The dependency is exact and checkable by hand: for all four nucleotides, the strong/weak bit is the exclusive-or of the purine/pyrimidine bit and the amino/keto bit. Writing the three bits in that order, adenine is $(1, 1, 0)$, cytosine is $(0, 1, 1)$, guanine is $(1, 0, 1)$, and uracil is $(0, 0, 0)$, and in each case the third coordinate is the sum of the first two modulo two.

Each nucleotide therefore carries two free bits, not three, and a codon carries six, which gives $2^6 = 64$. The corrected count reaches the same number by a route that is arithmetically consistent. We confirm separately that the resulting two-bit encoding is injective on the four nucleotides and that the induced map from the 64 codons onto $\{0, \ldots, 63\}$ is a bijection.

We note what this does and does not establish. It establishes that both sets have 64 elements and that a bijection exists. A bijection between two sets of the same finite size always exists, so the bijection alone carries no information about any relationship between the two systems.

4.8 Robustness

Three stress tests bound the results. First, the exact and idealized yarrow models are compared, and the state dependence that drives the main correction survives under both. Second, the analytic and numerical spectra are computed by independent routes and compared for both kernels and for the adjacency matrix. Third, quantities that are rational are compared exactly rather than within a tolerance, which removes floating-point accumulation as a source of agreement. The Monte-Carlo checks are the only sampled results, and they are reported with their trial count, seed, and observed deviation.

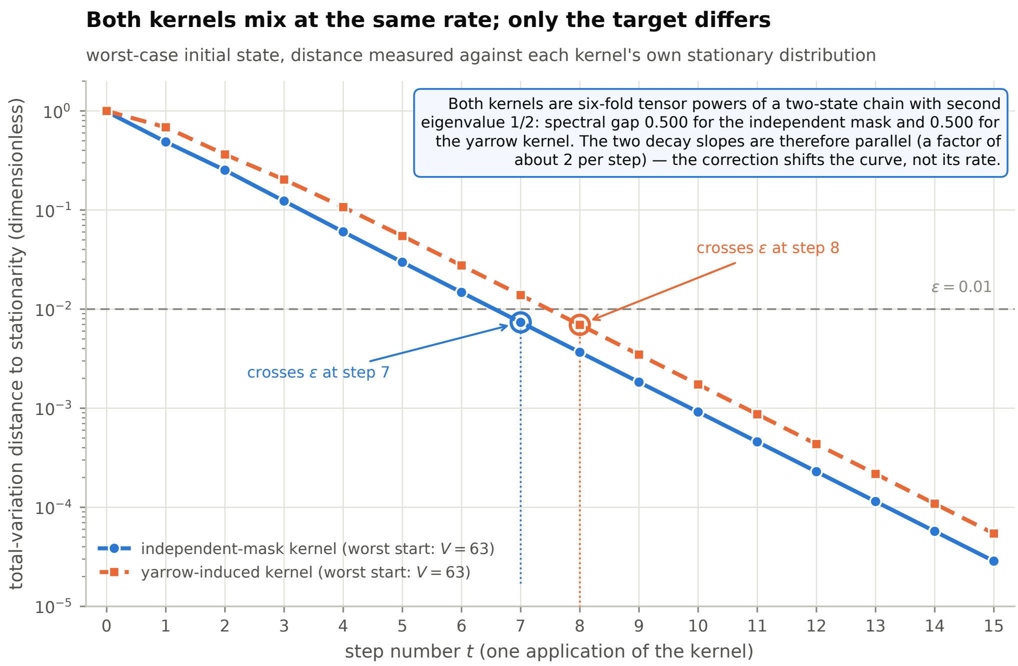

Figure 4: Total-variation distance to stationarity against step number, from the worst-case start. The two kernels share a spectral gap of $1/2$, so their decay rates agree even though their stationary distributions do not.

5 Discussion

5.1 What the corrections change

The corrections divide into three kinds, and they do not all have the same weight.

The clustering coefficient and the genetic-code dimension count are local errors. They do not propagate. Correcting the clustering coefficient to zero replaces a number with a structural fact, namely that the state graph is bipartite, which is more informative than the number it replaces. Correcting the dimension count removes an inconsistency while leaving the count of 64 intact.

The independence failure is different, because it propagates. It changes the stationary distribution from uniform to a product form with a 729-fold spread, it changes the mutual information, and it changes the entropy of the transformed hexagram. The reason it went unnoticed is instructive. The marginal change rate is exactly $1/4$, which is what the received analysis checked, and the cast hexagram is exactly uniform, which is what the entropy calculation needed. Both of those are true. The independence assumption is a third claim that neither of them implies, and it is false.

The substantive consequence is that the system has a direction. Under the kernel the divination procedure induces, repeated transformation is not a symmetric random walk on the hypercube. It is a biased walk with a preferred region, and the bias points toward yin. Whether that asymmetry is a designed feature of the procedure or an incidental consequence of the counting rules is not something the formalization can settle, and we do not claim it is.

5.2 Rival explanations

Three alternative readings deserve to be addressed before any of this is generalized.

The first is that the results are artifacts of the encoding. This is partly true and we have tried to bound it. The graph invariants are invariant under relabelling, so the clustering result does not depend on the encoding at all. The stationary distribution is stated in terms of the

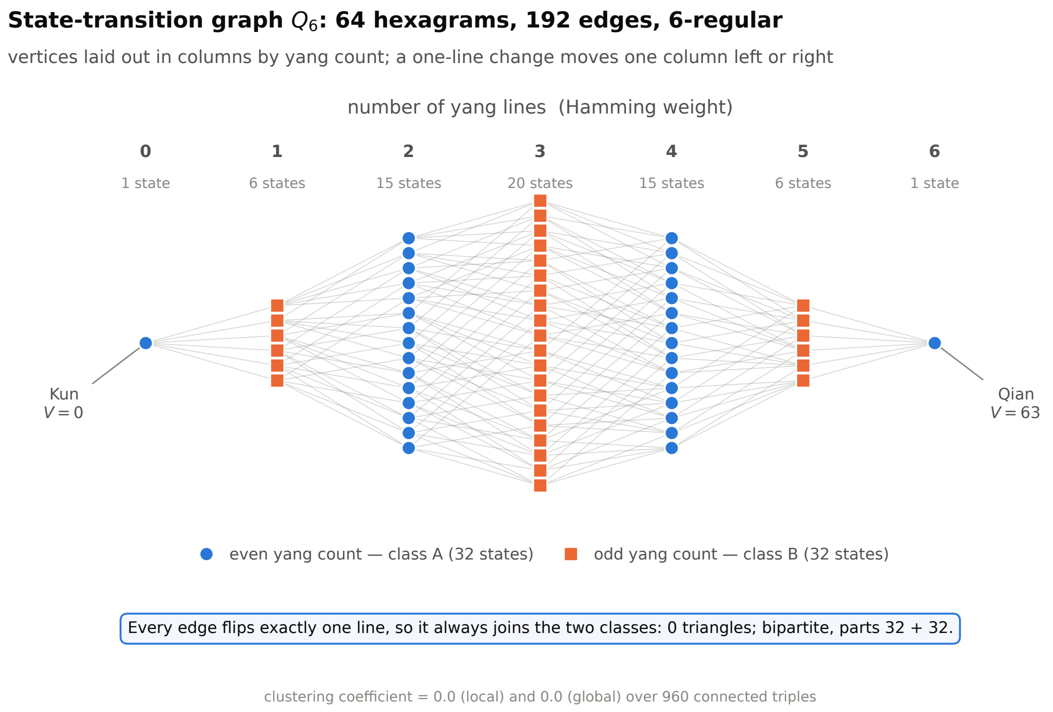

Figure 5: The state-transition graph, laid out in columns by number of yang lines. Every edge joins adjacent columns, so the graph is bipartite with parts of size 32 and 32 and contains no triangles. This is why its clustering coefficient is zero rather than 5/12.

number of yang lines rather than in terms of bit values, so it is also encoding-independent. The enumeration order is the one quantity that genuinely depends on convention, and we therefore report it under both conventions rather than picking one.

The second is that the yarrow-stalk procedure has attested variants, and the probabilities depend on which variant is modelled. This is correct. We model the three-operation procedure on 49 stalks under two explicit assumptions about how the pile is divided, and we report both. The coin-based method in common use has a different distribution over line states and would give different numbers. Our claim is about the procedure we model, not about divination in general.

The third is that the ordering of the arrangement admits more than one defensible reconstruction. We do not rely on any particular ordering for the algebraic, topological, or probabilistic results, all of which are properties of the set and its structure rather than of an enumeration. The ordering enters only in Section 3.1, where we report it under both conventions.

5.3 Relation to discrete computational models, and the limits of that relation

The system has features that recur in discrete models of physics. Its state space is finite, its transition rules are local in the sense that each generator affects one line, its states are generated recursively by bifurcation, and its deterministic dynamics are information preserving because exclusive-or is an involution. These are the features that motivate the broader literature on computable and discrete universe models, which represents dynamics as rewriting rules on discrete structures [13, 14], as quantum computation [15], or as causal sets [16], and which has been surveyed by Zenil [17].

We think the honest statement of the relation is a negative one, and we want to be specific about why.

The shared features listed above are shared by every system built from iterated binary

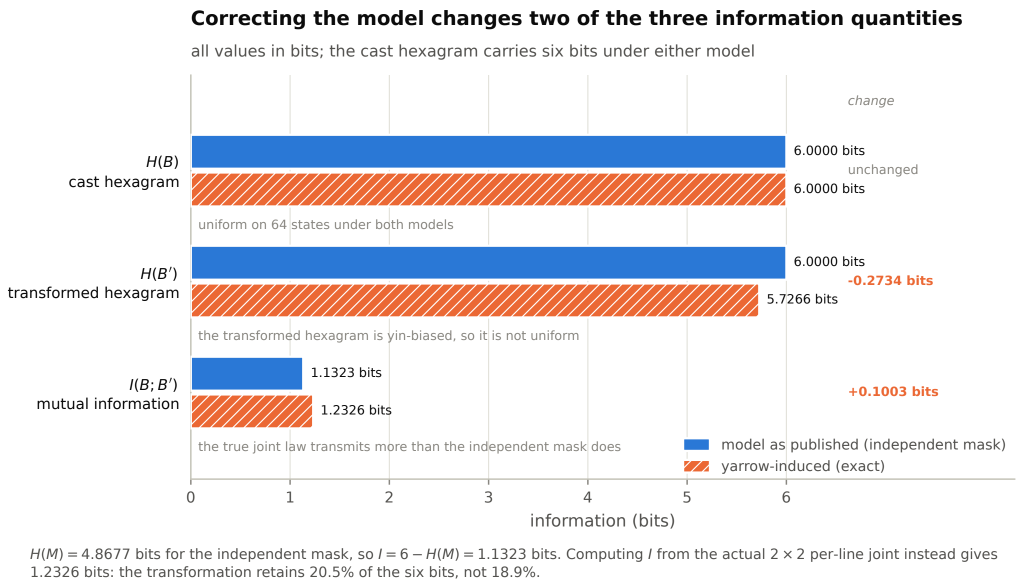

Figure 6: Information quantities in bits. The cast hexagram carries the full six bits. The transformed hexagram carries less, because it is yin-biased. The mutual information between them is higher than the independent-mask model predicts.

splitting. They are consequences of the construction, not evidence of a common origin. A finite state space with local involutive generators is what one gets by definition when one takes a finite abelian group of exponent two and its Cayley graph. Finding that structure in the hexagram arrangement is finding that the arrangement was built by repeated bifurcation, which is what the doubling generation says it was built by. Nothing further follows.

The scale gap is the second reason. The formalized system has 64 states. Any comparison to a physical theory is a comparison across many orders of magnitude, and the analogy does no work at that distance. Our system is a useful object precisely because it is small enough to enumerate exhaustively, which is the property that let us find the errors reported here. It is not small in a way that makes it a scale model of anything larger.

We therefore make no claim that the arrangement anticipated binary computation, the genetic code, or any model of discrete physics. The recurrence of powers of two across unrelated systems is what binary splitting produces, and it is not evidence of a relationship. The bijection between hexagrams and codons verified in Section 4.7 is a statement about two sets of size 64; a bijection between finite sets of equal size always exists, so its existence carries no information.

Predictions in this area, such as the error-minimizing structure of the genetic code [12] or observable signatures of discrete spacetime [18], are predictions of those independent hypotheses. They are not predictions of the present formalization, and we do not present them as such. Nor is the simulation argument [19] engaged by anything here; it is a probabilistic argument about posthuman computation, and a formal model of a symbolic system neither supports nor undermines it.

We also make no claim about the divinatory or interpretive tradition. The probability model describes the procedure by which line states are produced. It says nothing about what those line states are taken to mean.

5.4 Limitations

The formalization covers the Earlier Heaven arrangement and the three-operation yarrow procedure. It does not cover the King Wen sequence, the coin method, the line-position correspondences used in traditional interpretation, or any of the commentarial apparatus.

The probability results depend on a model of how the pile is divided. We report two such models and show the main conclusion is stable across them, but neither model is derived from empirical observation of how people actually divide a pile of stalks, and we are not aware of data that would settle it.

The Monte-Carlo results are estimates. They are reported with trial counts and seeds, and they agree with the exact computations, but they are not independent evidence for those computations since they simulate the same model.

Finally, a note on the citations. The twenty-two references circulating with the informal version of this analysis were each checked against a publisher record, a DOI, or an authoritative index. Nine were correct, twelve carried at least one wrong field, and one did not exist: an article attributed to *Studies of Zhouyi* 2006(3): 62–67 is absent from that issue's table of contents, and the description appears to belong to a different article in a different journal, which we cite instead [10]. The audit is recorded in the repository. Two of the comparative genetic-code references sit in venues from publishers widely listed as predatory, which we note so that readers can weigh them accordingly.

6 Conclusion

We have given a computational formalization of the Fuxi Earlier Heaven hexagram system and machine-checked the quantitative claims that the received account makes about it. The changing-line mechanism is bitwise exclusive-or, verified on all 4096 line-state assignments, and the transition structure is the elementary abelian group $\mathbb{Z}_2^6$ whose Cayley graph is the six-dimensional hypercube. Twenty-four of forty checked claims are confirmed and five are refined.

Seven are corrected. The state-transition graph is bipartite and therefore has a clustering coefficient of exactly zero rather than 5/12. The genetic-code dichotomies are dependent over GF(2), giving two free bits per nucleotide rather than three. Most consequentially, the yarrow-stalk procedure makes a yang line change three times as often as a yin line, so the mutation mask is not independent of the state. The kernel the procedure induces is asymmetric, its stationary distribution is the product form $\pi(h) \propto (3/4)^{6-y(h)}(1/4)^{y(h)}$ rather than uniform, and the mutual information between the cast and transformed hexagram is 1.233 bits rather than 1.132.

The scope of these results is the formal model. They establish properties of a six-bit state machine and of a probability model of one divination procedure. They do not establish anything about the interpretive tradition, and they do not support comparisons between this system and physical theory beyond the observation that both can be built by iterated binary splitting.

The methodological point generalizes further than the specific corrections do. Three errors persisted in an account of a 64-state system, which is small enough to enumerate completely on any computer. They persisted because the account was written rather than run. Formalizations of historical symbolic systems are worth executing, and the cost of doing so is low.

Code and data availability

The verification suite, the figure scripts, and the source of this manuscript are available at `https://github.com/AwareLiquid/The-Fuxi-Binary-System`. All results in this paper are regenerated by running `python verify_all.py` from the `code` directory. The suite requires only the Python standard library; NumPy is used solely for the numerical spectra that cross-check the analytic ones.

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